

AP EAMCET
Engineering Entrance Exam
25th April 2017- Shift 2

Mathematics

1. If $[\bullet]$ denote the greatest integer function and if $f: (5, 10) \rightarrow (7, 12)$ is a function defined by $f(x) = x + 2\left[\frac{x}{5}\right]$, then
 - A. $f^{-1}(x) = x - 1$
 - B. $f^{-1}(x) = x + 2$
 - C. $f^{-1}(x) = x - 2$
 - D. $f^{-1}(x)$ = does not exist
2. If $[\bullet]$ denotes the greatest integer function, then the domain and range of the function $f(x) = \frac{\sin[x]\pi + \tan[x]\pi}{1 + [x]^2 + [x]^4}$ are respectively
 - A. $\mathbb{R} - \{0\}, \mathbb{R} - \{0\}$
 - B. $\mathbb{R}^+, \{0\}$
 - C. $\mathbb{R}^+, \mathbb{R}$
 - D. $\mathbb{R}, \{0\}$
3. The least positive integer greater than 1 that divides $49^n + 16n - 1$ for all positive integers n is
 - A. 64
 - B. 49

- C.7
D.2
4. If A is a matrix of order 3 whose determinant is equal to 6, then $\det(\text{adj } A) =$
A.6
B.36
C.216
D.1
5. If A and B are square matrices of order 3, then $\left| (A - A^T) + (B - B^T) \right| =$
A. $2|A|$
B. $2|B|$
C. $2(|A| + |B|)$
D.0
6. The system of equations $x + y + z = 5$, $x + 2y + 3z = 9$ and $x + 3y + \lambda z = \mu$ has unique solution if
A. $\lambda = 5, \mu = 10$
B. $\lambda = 5, \mu \neq 10$
C. $\lambda \in \mathbb{R}, \mu \neq 5$
D. $\lambda \neq 5, \mu \in \mathbb{R}$
7. If a and c are complex numbers and b is a real number in the Argand plane, then the perpendicular distance from c to the line $a\bar{z} + \bar{a}z + b = 0$ is
A. $\frac{(a\bar{c} + \bar{a}c + b)}{2|a|}$
B. $\frac{(\overline{ac} + ac + b)}{2|a|}$

C. $\frac{(\bar{a}\bar{c} + \bar{a}\bar{c} + b)}{|a|}$

D. $\frac{(\bar{a} + b + \bar{c})}{2|a|}$

8. The real part of the complex number $= \frac{5+2i}{2-5i} - \frac{3-4i}{4+3i} - \frac{1}{i}$, is

A.2

B.0

C.3

D.4

9. If z , iz and $z + iz$ are the vertices of a triangle and if $|z| = 4$, then the area (in sq. units) of that triangle, is

A.4

B.8

C.16

D.32

10. $\sum_{n=1}^{20} \left[\sin\left(\frac{2n\pi}{21}\right) - i \cos\left(\frac{2n\pi}{21}\right) \right] =$

A.1

B.-1

C.i

D.-i

11. The integral value of k for which

$$x^2 - 2(4k-1)x + 15k^2 - 2k - 7 < 0 \text{ for all } x \in \mathbb{R} \text{ is}$$

A.2

B.3

C.1

D.4

12. If x_1, x_3 are the root of $Ax^2 - 4x + 1 = 0$ and x_2, x_4 are the roots of $Bx^2 - 6x + 1 = 0$ that x_1, x_2, x_3, x_4 are in harmonic progression, then

$$\frac{B+A}{B-A} =$$

A. $\frac{11}{5}$

B. $\frac{-11}{5}$

C. $\frac{5}{11}$

D. $\frac{-5}{11}$

13. If α and β are the root of $ax^2 + bx + c = 0$ then the roots of $ax^2 - bx(x-1) + c(x-1)^2 = 0$ are

A. $\frac{\alpha+1}{\alpha}, \frac{\beta+1}{\beta}$

B. $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

C. $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

D. $\frac{\alpha-1}{\alpha+1}, \frac{\beta-1}{\beta+1}$

14. If the roots of the equation $x^5 - 40x^4 - Px^3 - Rx - S = 0$ are in geometric progression and the sum of the reciprocals of the roots is 10, then ISI =

A. 8

B. 16

C. 32

D. 64

15. Aglose The number of all four digit numbers which do not have four distinct digits is

A. 4464

B. 4848

C. 4355

D. 4454

16. The number of ways of arranging the letters of the word BANANA so that the two Ns do not come together, is

A. 60

B. 80

C. 40

D. 120

17. The coefficient of x^{10} in the expansion of $(1+x^2-x^3)$ is

A. 506

B. 496

C. 486

D. 476

18. If $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, then $a_0 + a_2 + a_4 + \dots + a_{2n} =$

A. 3^n

B. $3^n + 1$

C. $\frac{3^n - 1}{2}$

D. $\frac{3^n + 1}{2}$

19. The values of x for which $\frac{x}{(x-1)^2(x-2)}$ has an expansion and

the coefficient of x^n in such expansion are respectively

A. $|x| < 1, 1 - n - \frac{1}{2^n}$

B. $|x| < 2, 1 - n - \frac{1}{2^n}$

C. $|x| < 1, 0$

D. $x \in \mathbb{R}, 1 - n - \frac{1}{2^n}$

20. The value of $\cot 70^\circ + 5 = 4 \cot 70^\circ$ is

A. $\frac{1}{\sqrt{3}}$

B. $\sqrt{3}$

C. $2\sqrt{3}$

D. $\frac{1}{2}$

21. $\frac{\sqrt{2} - \sin \alpha - \cos \alpha}{\sin \alpha - \cos \alpha}$

A. $\sin\left(\frac{\alpha}{2} - \frac{\pi}{8}\right)$

B. $\cos\left(\frac{\pi}{8} - \frac{\alpha}{2}\right)$

C. $\tan\left(\frac{\alpha}{2} - \frac{\pi}{8}\right)$

D. $\cot\left(\frac{\alpha}{2} - \frac{\pi}{8}\right)$

22. If $\sin(x - 3\alpha) + 3 \sin(x - \alpha) = 0$, then

A. $\tan x = \tan \alpha$

B. $\tan x = \tan^2 \alpha$

C. $\tan x = \tan^3 \alpha$

D. $\tan x = 3 \tan \alpha$

23. If $\cot \frac{x}{2} - \operatorname{cosec} \frac{x}{2} = \cot x$, then the value of x are

A. $2n\pi$

B. $4n\pi \pm \frac{2\pi}{3}$

C. $2n\pi \pm \frac{\pi}{3}$

D. $n\pi$

24. The number of solution of the equation $2\cos^{-1}x + \sin^{-1}x = \frac{11\pi}{6}$

A. 0

B. 1

C. 2

D. 3

25. If $\sinh^{-1}(2) + \sinh^{-1}(3) = \alpha$ then $\sinh \alpha =$

A. $2\sqrt{5} + 3\sqrt{10}$

B. $3\sqrt{10} + 3\sqrt{5}$

C. $2\sqrt{10} + 2\sqrt{5}$

D. $2\sqrt{10} + 3\sqrt{5}$

26. In $\triangle ABC$ if $a = 2b$ and $|A - B| = \frac{\pi}{3}$, then $\angle C =$

A. $\frac{\pi}{6}$

B. $\frac{\pi}{3}$

C. $\frac{\pi}{2}$

D. $\frac{\pi}{4}$

27. The base of a triangle is 80 and one of the base angles is 60° . If the sum of the lengths of the other two sides is 90. then the shortest side is of length

A. 15

B. 21

C. 19

D. 17

28. In $\triangle ABC$, if $r_1 = 3, r_2 = 10$ and $r_3 = 15$, then $R =$

- A. 5
- B. 12
- C. $\frac{13}{2}$
- D. 13

29. If $\vec{a}, \vec{b}$ are two non-parallel unit vectors and the vector $\alpha\vec{a} + \vec{b}$ bisects the internal angle between $\vec{a}$ and $\vec{b}$ then α is equal to

- A. 1
- B. $\frac{1}{2}$
- C. 2
- D. 3

30. If $\vec{OA} = 3\vec{i} + \vec{j} - \vec{k}, |\vec{AB}| = 2\sqrt{6}$ and the direction ratios of

$\vec{AB}$ are 1, -1, 2 then $|\vec{OB}| =$

- A. $\sqrt{41}$
- B. $\sqrt{35}$
- C. $\sqrt{26}$
- D. $\sqrt{55}$

31. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that

$|\vec{a}| = |\vec{c}| = 1, |\vec{b}| = 4, |\vec{b} \times \vec{c}| = \sqrt{15}$. if $\lambda\vec{a} = \vec{b} - 2\vec{c}$, then the value of λ is

- A. ± 1
- B. ± 2
- C. ± 3
- D. ± 4

32. let $\vec{u}, \vec{v}, \vec{w}$ be vectors such that $|\vec{u}|=1, |\vec{v}|=2, |\vec{w}|=3$. If the projection of $\vec{v}$ on $\vec{u}$ is equal to that of $\vec{w}$ on $\vec{u}$ and the vectors $\vec{v}, \vec{w}$ are perpendicular to each other, then $|\vec{u} - \vec{v} + \vec{w}| =$

- A. 2
- B. $\sqrt{7}$
- C. $\sqrt{14}$
- D. 14

33. If the $\vec{a}, \vec{b}, \vec{c}$ satisfy the condition $|\vec{a} - \vec{c}| = |\vec{b} - \vec{c}|$, then $(\vec{b} - \vec{a}) \cdot \left(\vec{c} - \frac{\vec{a} + \vec{b}}{2} \right) =$

- A. 0
- B. -1
- C. 1
- D. 2

34. For two given vectors $\vec{a}$ and $\vec{b}$ if the vector $\vec{A}, \vec{B}$ are

$\vec{A} + \vec{B} = \vec{a}, \vec{A} \times \vec{B} = \vec{b}$ and $\vec{A} \cdot \vec{a} = 1$ then $\vec{A} =$

- A. $\frac{(\vec{a} \times \vec{b}) + \vec{a}}{\vec{a}^2}$
- B. $\frac{(\vec{b} \times \vec{a}) + \vec{a}(\vec{a}^2 - 1)}{\vec{a}^2}$
- C. $\frac{\vec{a}(\vec{a}^2 - 1) + \vec{b}(\vec{b}^2 - 1)}{\vec{a}^2 + \vec{b}^2}$
- D. $\frac{(\vec{a} \times \vec{b}) + \vec{b}}{\vec{b}^2}$

35. The mean of numbers a, b, 8, 5, 10 is 6 and their variance is

680 Then $\tan^{-1} \frac{1}{a} + \tan^{-1} \frac{1}{b} =$

- A. $\tan^{-1} \frac{7}{12}$

B. $\tan^{-1}\left(-\frac{7}{11}\right)$

C. $\tan^{-1}\frac{11}{7}$

D. $\tan^{-1}\frac{7}{11}$

36. If the coefficients of variation of two distributions are 60 and 70 and their standard deviations are 21 and 16 respectively, then their arithmetic means are respectively

A. 35, 22.85

B. 32, 25.85

C. 35, 28.25

D. 35, 25.25

37. If E_1, E_2 are two events of a sample space such that

$$P(E_1) = \frac{1}{4}, P(E_2 | E_1) = \frac{1}{2}, P(E_1 | E_2) = \frac{1}{4}, \text{ then } P(\overline{E_1} | \overline{E_2}) =$$

A. $\frac{1}{3}$

B. $\frac{1}{4}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

38. If A and B are any two events such that

$$P(\overline{A \cup B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4} \text{ and } P(\overline{A}) = \frac{1}{4} \text{ then the events A and B are}$$

A. Equally likely but not independent

B. Equally likely and mutually exclusive

C. Mutually exclusive and independent

D. Independent but not equally likely

39. For a random variable X if $P(X=k) = \frac{(k+1)a}{3^k}$ for $k=0, 1, 2, \dots$

Then a=

A. $\frac{2}{3}$

B. $\frac{4}{9}$

C. $\frac{8}{27}$

D. $\frac{16}{81}$

40. On an average if one out of 100 electric bulbs produced by a Company is found to be defective, then the probability that there are at least two defective bulbs in a consignment of 600 bulbs. Is

A. $1 - 7e^{-6}$

B. $1 - 6e^{-6}$

C. $1 - 6e^{-1}$

D. $1 - 8e^{-4}$

41. If $A = (1, 0)$ and $B = (-a, 0)$ then the locus of a point P such that

$$PA^2 - PB^2 = a^2 \text{ is}$$

A. a circle

B. an ellipse

C. a hyperbola

D. a straight line

42. The angle through which the coordinate axes are to be rotated to remove the Ay term in the equation $x^2 + 2xy - y^2 = 0$ is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{3}$

C. $\frac{\pi}{8}$

D. $\frac{\pi}{6}$

43. The point P (a, b) lies on the straight line $3x + 2y = 13$ and the point Q (b, a) lies on the straight line $4x - y = 5$. Then the equation of the line PQ is

A. $x + y = 7$

B. $x + y = 5$

C. $x + y = 2$

D. $x + y = 21$

44. If three non-zero real numbers a, b, c are in harmonic progression, then the straight lines $\frac{x}{a} + \frac{y}{b} + \frac{2}{c} = 0$ are concurrent at the point

A. (1, -2)

B. (-2, 4)

C. (4, -2)

D. (-2, -4)

45. The incentre of the triangle with vertices

$A(1, \sqrt{3}), B(0, 0)$ and $C(2, 0)$ is

A. $\left(1, \frac{\sqrt{3}}{2}\right)$

B. $\left(\frac{2}{3}, \frac{1}{\sqrt{3}}\right)$

C. $\left(\frac{2}{3}, \frac{\sqrt{3}}{2}\right)$

D. $\left(1, \frac{1}{\sqrt{3}}\right)$

46. The distance between the lines represented by

$$x^2 + 2xy + y^2 - 8mx - 8my - 9m^2 = 0$$

A. $2\sqrt{5}m$

B. 0

C. $\frac{5m}{\sqrt{2}}$

D. $5\sqrt{2}m$

47. If $\frac{x^2}{a} + \frac{2xy}{h} + \frac{y^2}{b} = 0$ represents a pair of straight lines such that the slope of one of the line is twice the other, then $\frac{ab}{h^2} =$

A. $\frac{9}{8}$

B. $\frac{8}{9}$

C. $\frac{2\sqrt{2}}{3}$

D. $\frac{3}{2\sqrt{2}}$

48. The duce sides of a triangle are given by

$$(x^2 + 7xy + 2y^2)(y - 1) = 0. \text{ Then the centroid of that triangle is}$$

A. $\left(\frac{2}{3}, 0\right)$

B. $\left(\frac{7}{3}, \frac{2}{3}\right)$

C. $\left(-\frac{7}{3}, \frac{2}{3}\right)$

D. $\left(\frac{1}{3}, \frac{4}{3}\right)$

49. Let α, β be the roots of :

$$x^2 + 5x + 6 = 0 \text{ and } \gamma, \delta \text{ be the roots of } y^2 + 6y + 7 = 0 \text{ Then the equation of}$$

the circle with (α, γ) and (β, δ) as the extremities of a diameter is

A. $x^2 + y^2 + 5x + 6y + 10 = 0$

B. $x^2 + y^2 + 5x + 6y + 11 = 0$

C. $x^2 + y^2 + 5x + 6y + 13 = 0$

D. $x^2 + y^2 + 5x + 6y + 12 = 0$

50. If A(2, c) and B(d, 2) are two points such that the polar of one point with respect to the circle $x^2 + y^2 = 16$ passes through the other, then $c + d =$

A. 4

B. 6

C. 8

D. 10

51. The number of common tangents to the circles

$x^2 + y^2 + 2x + 8y - 23 = 0$ and $x^2 + y^2 - 4x - 10y + 19 = 0$, is

A. 1

B. 2

C. 3

D. 4

52. If the circles $x^2 + y^2 + 2hx + 2ky = 0$ and $x^2 + y^2 + 2h'x + 2k'y = 0$ touch each other. Then $\frac{h'k}{hk'} =$

A. 0

B. 1

C. 2

D. -1

53. If the circles $x^2 + y^2 + kx + 4y + 2 = 0$ and $2(x^2 + y^2) - 4x - 3y + k = 0$ cut orthogonally. then $k =$

A. $\frac{-10}{3}$

B. $\frac{10}{3}$

C. $\frac{5}{3}$

D. $\frac{-5}{3}$

54. The equation of the circle passing through the points of intersection of the circle $x^2 + y^2 - 2px = 0$ and $x^2 + y^2 - 2qy = 0$ and having its center on $\frac{x}{p} - \frac{y}{q} = 2$, is

A. $x^2 + y^2 + 3px + qy = 0$

B. $x^2 + y^2 + px - qy = 0$

C. $x^2 + y^2 - 3px + qy = 0$

D. $x^2 + y^2 - 4px + 2qy = 0$

55. If the normal at the point t_1 (i.e., at $(at_1^2, 2at_1)$) on $y^2 = 4ax$ meets the parabola again at the point t_2 , then $t_1 t_2 =$

A. $-2 - t_1^2$

B. -2

C. $-t_1^2 + 2$

D. $t_1^2 + 3$

56. The equation of the common tangent to the parabolas $y^2 = 32x$ and $x^2 = 256y$ is

A. $x + 2y - 32 = 0$

B. $x + 2y + 32 = 0$

C. $2x + 2y - 32 = 0$

D. $2x + 2y + 32 = 0$

57. The equation $\frac{x^2}{2-r} + \frac{y^2}{r-5} + 1 = 0$ represents an ellipse if

A. $r > 2$

B. $r > 5$

C. $2 > r < 5$

D. $r > 2$ or $r > 5$

58. The equation of the locus of the foot of the perpendicular drawn from the centre of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ to any tangent of the ellipse is

A. $(x^2 + y^2)^2 = a^2x^2 + b^2y^2$

B. $(x^2 - y^2)^2 = a^2x^2 + b^2y^2$

C. $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$

D. $(x^2 - y^2)^2 = a^2x^2 - b^2y^2$

59. The product of the perpendicular distances drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to its asymptotes is

A. $\frac{13}{36}$

B. $\frac{13}{5}$

C. $\frac{36}{13}$

D. $\frac{36}{5}$

60. The points $(2, 3, 5)$, $(-1, 5, -1)$ and $(4, -3, 2)$ form

A. a right angled but not an isosceles triangle

B. an isosceles but not a right angled triangle

C. an equilateral triangle

D. an isosceles right angled triangle

61. If the direction cosines l, m, n of two lines are satisfying the relations $l + m + n = 0, lm = 0$ then the angle between those two lines is

A. 0

B. $\frac{\pi}{2}$

C. $\frac{\pi}{4}$

D. $\frac{\pi}{3}$

62. If $(2, -3, 6)$ is the foot of the perpendicular drawn from the origin to a plane, then the equation of that plane is

A. $2x + 4y - 3z - 29 = 0$

B. $2x - 5y + 8z + 98 = 0$

C. $2x + 4y + 3z + 29 = 0$

D. $2x - 3y + 6z - 49 = 0$

63. If $f(x) = \begin{cases} \frac{1 - \cos ax}{x \sin x}, & (x \neq 0) \\ \frac{1}{2}, & (x = 0) \end{cases}$ and f is continuous at $x = 0$, then $a^2 =$

A. 1

B. 2

C. 3

D. 4

64. If $f(9) = 9$ and $f'(9) = 4$, then $\lim_{x \rightarrow 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3} =$

A. 2

B. 3

C. 9

D.4

65. The points at which the function $f(x) = |x - 0.5| + |x - 1| + \tan x$ does not have a derivative in the interval $(0, 2)$ is

A. 0.5, 1.2

B. $0.5, \frac{\pi}{2}, 0$

C. $1, \frac{\pi}{2}, 2$

D. $0.5, 1, \frac{\pi}{2}$

66. If $y = \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right)$ for $0 < |x| < 1$, then $\frac{dy}{dx} =$

A. $\frac{x}{\sqrt{1-x^2}}$

B. $\frac{x^2}{\sqrt{1-x^4}}$

C. $\frac{\sqrt{1+x^2}}{\sqrt{1-x^4}}$

D. $\frac{-x}{\sqrt{1-x^4}}$

67. If $y = t^2 + t^3$ and $x = t - t^4$ then the value of $\frac{d^2y}{dx^2}$ at $t = 1$ is equal to

A. $\frac{2}{3}$

B. $\frac{-2}{3}$

C. $\frac{4}{3}$

D. $\frac{-4}{3}$

68. If the normal to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ makes an angle ϕ with the X - axis, then the equation of that normal is

A. $y - \cos^2 \phi = x \tan \phi - a^2 \sin^2 \phi$

B. $y \cos \phi - x \sin \phi = a \cos 2\phi$

C. $y \cos \phi - x \sin \phi = a \cos^2 \phi$

D. $y + a \sin^2 \phi = x \cos \phi - a \sin 2\phi$

69. If the rate of change of x is more than the rate of change of y on the curve $x^3 = 12y$, ($x > 0$) then x lies in the interval

A. $(-2, 0) \cup (0, 2)$

B. $(-2, 2)$

C. $[-3, 3]$

D. $(0, 2)$

70. If $0 < x < \frac{\pi}{2}$ then the maximum area (in sq. units) of the triangle whose vertices are $(0, 0)$, $(x, \cos x)$ and $(\sin^3 x, 0)$ is

A. $\frac{3\sqrt{3}}{32}$

B. $\frac{7\sqrt{3}}{32}$

C. $\frac{5\sqrt{3}}{32}$

D. $\frac{3\sqrt{3}}{16}$

71. Suppose $f(x)$ is twice differentiable in the interval $[1, 3]$ and $f(1) = f(3)$. If I , then for all x in $[1, 3]$, which one of the following is true?

A. $|f'(x)| \geq 1$

B. $-4 < f'(x) < 4$

C. $|f'(x)| > 2$

D. $-3 \leq f'(x) \leq 3$

72. If $0 < a < 1$, then $\int \frac{dx}{1 - 2a \cos x + a^2} =$

A. $\frac{1}{1-a^2} \tan^{-1} \left[\frac{1+a}{1-a} \tan \frac{x}{2} \right] + c$

B. $\frac{1}{1+a^2} \tan^{-1} \left[\frac{1-a}{1+a} \tan \frac{x}{2} \right] + c$

C. $\frac{2}{1-a^2} \tan^{-1} \left[\frac{1+a}{1-a} \tan \frac{x}{2} \right] + c$

D. $\frac{2}{1+a^2} \tan^{-1} \left[\frac{1-a^2}{1+a^2} \tan \frac{x}{2} \right] + c$

73. If $\int \sin^5 x dx = \frac{-\cos^5 x}{5} + a \cos^3 x + b \cos x + c$, then $a + b =$

A. $-\frac{1}{3}$

B. 0

C. $\frac{1}{3}$

D. $\frac{2}{3}$

74. $\int \frac{6x+5}{\sqrt{6+x-2x^2}} dx =$

A. $-3\sqrt{6+x-2x^2} + \frac{13}{2\sqrt{2}} \sin^{-1} \left(\frac{4x-1}{7} \right) + c$

B. $-3\sqrt{6+x-2x^2} + \frac{13}{\sqrt{2}} \sinh^{-1} \left(\frac{4x-1}{7} \right) + c$

C. $-3\sqrt{6+x-2x^2} + \frac{13}{2\sqrt{2}} \sinh^{-1} \left(\frac{4x-1}{7} \right) + c$

D. $3\sqrt{6+x-2x^2} + \frac{13}{2\sqrt{2}} \cos^{-1} \left(\frac{4x-1}{7} \right) + c$

75. $\int_{-1}^{\frac{3}{2}} |x \sin \pi x| dx$

A. $\frac{2}{\pi} + \frac{1}{\pi^2}$

B. $\frac{3}{\pi} + \frac{1}{\pi^2}$

C. $\frac{4}{\pi} + \frac{1}{\pi^2}$

D. $\frac{1}{\pi} + \frac{2}{\pi^2}$

76. $\lim_{n \rightarrow \infty} \left[\frac{1^2}{1^3 + n^3} + \frac{2^2}{2^3 + n^3} + \dots + \frac{r^2}{r^3 + n^3} + \dots \frac{1}{2n} \right] =$

A. $\sqrt[4]{\log 2}$

B. $\log \sqrt[3]{2}$

C. $\sqrt{\log 2}$

D. $\log \sqrt[3]{3}$

77. The area of the region that is common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 6ax$ is

A. $\frac{4a^2}{3} (4\pi + \sqrt{3})$

B. $\frac{2a^2}{3} (3\pi + \sqrt{3})$

C. $\frac{4a^2}{3} (2\pi + \sqrt{2})$

D. $\frac{2a^2}{3} (2\pi + \sqrt{3})$

78. $\int_0^\pi x \sin^7 x \cos^6 x dx =$

A. $\frac{8\pi}{1002}$

B. $\frac{18\pi}{1003}$

C. $\frac{16\pi}{3003}$

D. $\frac{6\pi}{3003}$

79. The general solution of the differential equation $(x-y)^2 \frac{dy}{dx} = a^2$

is

A. $y = a \log \left(\frac{x-y+1}{x+y-1} \right) + c$

B. $y = \frac{a}{2} \log \left(\frac{x-y-a}{x-y+a} \right) + c$

C. $y = \frac{a}{2} \log \left(\frac{x+y-a}{x-y-a} \right) + c$

D. $y = \frac{a}{4} \log \left(\frac{x+y-a}{x+y+a} \right) + c$

80. The solution of the differential equation $2 \frac{dy}{dx} - \frac{y}{x} = \frac{y^2}{x^2}$, given

that $y = 2$ when $x = 1$, is

A. $2y = \frac{x}{1+\sqrt{x}}$

B. $y = \frac{2x}{2-\sqrt{x}}$

C. $y = \frac{x}{2(1+\sqrt{x})}$

D. $y = \frac{2x}{1+\sqrt{x}}$